TestLink

# Test Specification

## Test Project: UniShare
## Test Suite: UniShare - Rawa Dara - Sharing, Articles & Notifications (Iteration 4)

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2026 Group UniShare

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## Scope

UniShare - university course sharing platform (Next.js / tRPC / Prisma). Group project for SECJ3343 Project 1 (Aland, Karoz, Parwar, Rawa). Source: github.com/Seganation/unishare

# 1.Test Suite : Rawa Dara - Sharing, Articles & Notifications (Iteration 4)

**Tester:** Rawa Dara
**Software under test:** UniShare (github.com/Seganation/unishare)
**Test basis:** UniShare_Iteration_4.md use cases UC-S4-001, UC-S4-003, UC-S4-004 and the tRPC routers course.ts, resource.ts, article.ts, notification.ts.
**Techniques:** Equivalence Partitioning (EP), Boundary Value Analysis (BVA), Decision Table (DT), State Transition (ST), Use Case / scenario testing (UC), Error Guessing (EG).

**Common test data**
Student A - Sarah Ahmed (sarah@student.utm.my), course owner, APPROVED, Faculty of Computing
Student B - John Doe (john@student.utm.my), APPROVED, Faculty of Computing (same faculty as A)
Student C - Michael Chen (michael@student.utm.my), APPROVED, Faculty of Engineering (different faculty)
Student D - Alice Wong (alice@student.utm.my), role PENDING (not approved)
Seed data: prisma/seed.ts (all seeded passwords documented there).
Course: "SECJ3343 Software Quality Assurance" created by Student A with one resource card "Lecture Slides" (file uploads allowed) holding file week1.pdf.

## 1.1.Test Suite : TS-R1 Share Course (UC-S4-001)

Owner invites students to a course. Techniques: EP, DT, EG.

| **Test Case US-1: Owner invites same-faculty student as VIEWER [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Owner invites same-faculty student as VIEWER.<br><br>**Test design technique:** Use case (main flow), EP valid class<br><br>**Requirements:** REQ-SHR-03, REQ-SHR-07, REQ-NOT-01 | | |
| Preconditions:<br><br>Student A logged in and owns course SECJ3343. Student B exists, same faculty, not yet invited. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open Courses > SECJ3343 Software Quality Assurance > Collaborators. | Collaborators page shows 'Manage who has access to SECJ3343 Software Quality Assurance'. |
| 2 | Enter Student B's email in the invite field (colleague@university.edu placeholder). | Email accepted in the field. |
| 3 | Select permission 'Viewer' and click Invite. | Success message shown; Student B listed with role Viewer and status Pending. |
| 4 | Log in as Student B and open the notification bell. | Notification 'New course invitation': 'Student A invited you to collaborate on SECJ3343 Software Quality Assurance'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-SHR-03: Invitation carries a permission level [Version : 1]<br>REQ-SHR-07: Owner manages collaborators [Version : 1]<br>REQ-NOT-01: Event notifications [Version : 1] | |
| Keywords: | None | |

| **Test Case US-2: Owner invites same-faculty student as CONTRIBUTOR [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Owner invites same-faculty student as CONTRIBUTOR.<br><br>**Test design technique:** EP (role = CONTRIBUTOR partition)<br><br>**Requirements:** REQ-SHR-03, REQ-SHR-07 | | |
| Preconditions:<br><br>Student A owns SECJ3343. Student B has no invitation for this course. | | |
| #: | Step actions: | Expected Results: |
| 1 | On the Collaborators page enter Student B's email, select 'Contributor', click Invite. | Invitation created with role Contributor, status Pending. |
| 2 | Log in as Student B, open Settings > Pending Invitations. | Invitation for SECJ3343 is listed showing permission Contributor and owner Student A. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-SHR-03: Invitation carries a permission level [Version : 1]<br>REQ-SHR-07: Owner manages collaborators [Version : 1] | |
| Keywords: | None | |

| **Test Case US-3: Non-owner cannot invite collaborators [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Non-owner cannot invite collaborators.<br><br>**Test design technique:** DT (actor = collaborator/other student), negative<br><br>**Requirements:** REQ-SHR-01 | | |
| Preconditions:<br><br>Student B is an ACCEPTED CONTRIBUTOR of SECJ3343 (owner Student A). Student C exists. | | |
| #: | Step actions: | Expected Results: |
| 1 | Log in as Student B, open SECJ3343 > Collaborators. | Invite controls are hidden or disabled for a non-owner. |
| 2 | As Student B send an invite request for Student C to SECJ3343 (e.g. via the API course.inviteCollaborator). | Request rejected with FORBIDDEN 'Only the owner can invite collaborators'; no invitation created. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-SHR-01: Only the course owner can invite collaborators [Version : 1] | |
| Keywords: | None | |

| **Test Case US-4: Owner cannot invite himself [Version : 1]** | |
|---|---|
| Author: | rawa |

| | | |
|---|---|---|
| Summary:<br><br>Owner cannot invite himself.<br><br>**Test design technique:** EG / EP invalid class (invitee = owner)<br><br>**Requirements:** REQ-SHR-02 | | |
| Preconditions:<br><br>Student A logged in, owns SECJ3343. | | |
| #: | Step actions: | Expected Results: |
| 1 | On the Collaborators page enter Student A's own email, choose Viewer, click Invite. | Error 'You can't invite yourself'; no collaborator row added. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-SHR-02: Invalid invitations are rejected [Version : 1] | |
| Keywords: | None | |

| | | |
|---|---|---|
| **Test Case US-5: Duplicate invitation to the same student is rejected [Version : 1]** | | |
| Author: | rawa | |
| Summary:<br><br>Duplicate invitation to the same student is rejected.<br><br>**Test design technique:** EP invalid class (already invited), DT<br><br>**Requirements:** REQ-SHR-02 | | |
| Preconditions:<br><br>Student B already has a PENDING (repeat with ACCEPTED) invitation to SECJ3343. | | |
| #: | Step actions: | Expected Results: |
| 1 | Invite Student B again to SECJ3343 as Contributor. | Error 'User already invited'; existing invitation unchanged (role and status not modified). |
| 2 | Repeat with Student B's invitation in ACCEPTED status. | Same error 'User already invited'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-SHR-02: Invalid invitations are rejected [Version : 1] | |
| Keywords: | None | |

| | | |
|---|---|---|
| **Test Case US-6: Invite to a course that does not exist [Version : 1]** | | |
| Author: | rawa | |
| Summary:<br><br>Invite to a course that does not exist.<br><br>**Test design technique:** EG, EP invalid course id<br><br>**Requirements:** REQ-SHR-01 | | |
| Preconditions:<br><br>Student A logged in. | | |

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Call course.inviteCollaborator with courseId 'does-not-exist', a valid user id and role VIEWER. | NOT_FOUND 'Course not found'; nothing stored. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Low | |
| Requirements | REQ-SHR-01: Only the course owner can invite collaborators [Version : 1] | |
| Keywords: | None | |

| **Test Case US-7: Invitation to a student from a different faculty is blocked [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Invitation to a student from a different faculty is blocked.<br><br>**Test design technique:** EP (same faculty / different faculty partitions)<br><br>**Requirements:** REQ-SHR-08 | | |
| Preconditions:<br><br>Student A (Faculty of Computing) owns SECJ3343. Student C is in Faculty of Engineering. | | |
| #: | Step actions: | Expected Results: |
| 1 | Search/enter Student C's email on the Collaborators page and click Invite as Viewer. | System refuses the invitation with a message that only students of the same faculty can be invited; no invitation or notification is created. |
| 2 | Invite Student B (same faculty) the same way. | Invitation created (valid partition) - confirms the rule is faculty based. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-SHR-08: Faculty-restricted sharing [Version : 1] | |
| Keywords: | None | |

| **Test Case US-8: Owner changes collaborator permission and removes collaborator [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Owner changes collaborator permission and removes collaborator.<br><br>**Test design technique:** ST (VIEWER -> CONTRIBUTOR -> removed)<br><br>**Requirements:** REQ-SHR-07 | | |
| Preconditions:<br><br>Student B is ACCEPTED VIEWER on SECJ3343. | | |
| #: | Step actions: | Expected Results: |
| 1 | As Student A open Collaborators, change Student B's role to Contributor and save. | Student B shown as Contributor; Student B can now add a resource (see TS-R3). |
| 2 | Change Student B back to Viewer. | Role shown as Viewer; Student B can no longer add resources. |

| | | |
|---|---|---|
| 3 | Click Remove for Student B and confirm. | Student B disappears from the list; SECJ3343 no longer appears in Student B's courses. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-SHR-07: Owner manages collaborators [Version : 1] | |
| Keywords: | None | |

## 1.2.Test Suite : TS-R2 Respond to Course Invitation

Invitee accepts / declines. Techniques: ST (PENDING -> ACCEPTED/REJECTED), DT.

| **Test Case US-9: Invitee accepts a pending course invitation [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Invitee accepts a pending course invitation.<br><br>**Test design technique:** ST (PENDING -> ACCEPTED), use case main flow<br><br>**Requirements:** REQ-SHR-04, REQ-NOT-01 | | |
| Preconditions:<br><br>Student B has a PENDING VIEWER invitation to SECJ3343 from Student A. | | |
| #: | Step actions: | Expected Results: |
| 1 | Log in as Student B, open Settings > Pending Invitations. | Invitation listed with course title, owner name and permission Viewer. |
| 2 | Click Accept. | Invitation removed from Pending Invitations; success message shown. |
| 3 | Open Courses. | SECJ3343 Software Quality Assurance is listed as a shared course with role Viewer. |
| 4 | Log in as Student A and open the bell. | Notification 'Invitation accepted': 'Student B accepted your invitation to SECJ3343 Software Quality Assurance'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1]<br>REQ-NOT-01: Event notifications [Version : 1] | |
| Keywords: | None | |

| **Test Case US-10: Invitee declines a pending course invitation [Version : 1]** | |
|---|---|
| Author: | rawa |
| Summary:<br><br>Invitee declines a pending course invitation.<br><br>**Test design technique:** ST (PENDING -> REJECTED)<br><br>**Requirements:** REQ-SHR-04, REQ-NOT-01 | |

| Preconditions:<br><br>Student B has a PENDING invitation to SECJ3343. | | |
|---|---|---|
| #: | Step actions: | Expected Results: |
| 1 | As Student B open Settings > Pending Invitations and click Decline. | Invitation removed from the list; when none remain 'No pending invitations' is shown. |
| 2 | Open Courses as Student B. | SECJ3343 is NOT listed. |
| 3 | As Student A open the bell. | Notification 'Invitation declined': 'Student B declined your invitation to SECJ3343 Software Quality Assurance'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1]<br>REQ-NOT-01: Event notifications [Version : 1] | |
| Keywords: | None | |

| **Test Case US-11: Answered invitation cannot be accepted or declined again [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Answered invitation cannot be accepted or declined again.<br><br>**Test design technique:** ST (invalid transition from ACCEPTED/REJECTED), EG<br><br>**Requirements:** REQ-SHR-04 | | |
| Preconditions:<br><br>Student B's invitation to SECJ3343 is already ACCEPTED. | | |
| #: | Step actions: | Expected Results: |
| 1 | Call course.acceptInvitation for SECJ3343 again as Student B. | Request fails (record not found for status PENDING); status stays ACCEPTED, joinedAt unchanged, no new notification to owner. |
| 2 | Call course.rejectInvitation for SECJ3343 as Student B. | Request fails; status stays ACCEPTED. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1] | |
| Keywords: | None | |

## 1.3.Test Suite : TS-R3 Shared Course Permissions (Viewer / Contributor)

Decision table: role {Owner, Contributor, Viewer} x action {view, download, add, upload, delete resource, delete course}.

| **Test Case US-12: VIEWER can view and download but cannot modify [Version : 1]** | |
|---|---|
| Author: | rawa |
| Summary:<br><br>VIEWER can view and download but cannot modify.<br><br>**Test design technique:** DT (role = VIEWER column) | |

**Requirements:** REQ-SHR-05

Preconditions:

Student B is ACCEPTED VIEWER on SECJ3343; card 'Lecture Slides' holds week1.pdf.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Log in as Student B and open SECJ3343. | Course and all resource cards are displayed. |
| 2 | Download week1.pdf from 'Lecture Slides'. | File downloads successfully. |
| 3 | Look for Add resource / Upload / Edit / Delete controls. | Controls are not shown (read-only view). |
| 4 | Attempt resource.create on SECJ3343 via API as Student B. | FORBIDDEN 'Only course owner and contributors can create resources'. |
| 5 | Attempt to upload a file to 'Lecture Slides' via API as Student B. | FORBIDDEN 'You don't have permission to add files to this resource'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-SHR-05: VIEWER has read-only access [Version : 1] |
| Keywords: | None |

**Test Case US-13: CONTRIBUTOR can add resources and upload files [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

CONTRIBUTOR can add resources and upload files.

**Test design technique:** DT (role = CONTRIBUTOR, allowed actions)

**Requirements:** REQ-SHR-06

Preconditions:

Student B is ACCEPTED CONTRIBUTOR on SECJ3343.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | As Student B open SECJ3343 and add resource card 'Tutorial 1'. | Card created and visible to Student A as well. |
| 2 | Upload tutorial1.pdf (2 MB) into 'Lecture Slides'. | File uploaded and listed in the card. |
| 3 | Edit the title of 'Tutorial 1' to 'Tutorial 1 - BVA'. | Change saved. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-SHR-06: CONTRIBUTOR can add but not delete [Version : 1] |
| Keywords: | None |

**Test Case US-14: CONTRIBUTOR cannot delete resources or the course [Version : 1]**

| | |
|---|---|
| Author: | rawa |

| | | |
|---|---|---|
| Summary:<br><br>CONTRIBUTOR cannot delete resources or the course.<br><br>**Test design technique:** DT (role = CONTRIBUTOR, forbidden actions)<br><br>**Requirements:** REQ-SHR-06 | | |
| Preconditions:<br><br>Student B is ACCEPTED CONTRIBUTOR on SECJ3343. | | |
| #: | Step actions: | Expected Results: |
| 1 | As Student B try to delete resource 'Lecture Slides'. | Delete control hidden; API call returns FORBIDDEN 'Only the course owner can delete resources'. Resource still exists. |
| 2 | As Student B open SECJ3343 > Settings and try 'Delete course'. | Not allowed; course still exists. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-SHR-06: CONTRIBUTOR can add but not delete [Version : 1] | |
| Keywords: | None | |

| **Test Case US-15: Non-collaborator and declined user cannot open a private course [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Non-collaborator and declined user cannot open a private course.<br><br>**Test design technique:** EP (no relation / REJECTED / PENDING partitions)<br><br>**Requirements:** REQ-SHR-04, REQ-SHR-05 | | |
| Preconditions:<br><br>Student C has no relation to SECJ3343; Student B's invitation is REJECTED. | | |
| #: | Step actions: | Expected Results: |
| 1 | As Student C open /courses/<SECJ3343 id> directly. | Access denied / course not found; no resources shown. |
| 2 | As Student B (rejected) open the same URL. | Access denied. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1]<br>REQ-SHR-05: VIEWER has read-only access [Version : 1] | |
| Keywords: | None | |

## 1.4.Test Suite : TS-R4 Create & Edit Article (UC-S4-003)

My Articles > New Article / Edit. Techniques: BVA on title (1..200) and excerpt (0..500), EP on cover URL.

| **Test Case US-16: Create article saved as draft (valid data) [Version : 1]** | |
|---|---|
| Author: | rawa |
| Summary:<br><br>Create article saved as draft (valid data). | |

**Test design technique:** Use case main flow, EP valid

**Requirements:** REQ-ART-02, REQ-ART-10

Preconditions:

Student A logged in.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Open My Articles > New Article. | Form shows Title*, Excerpt (Optional), Cover Image (Optional), Tags (Optional), Content. |
| 2 | Title 'Intro to Boundary Value Analysis', Excerpt 'Short guide to BVA', Tags 'Testing, QA', some content; click Save Draft. | Toast 'Article saved as draft'. |
| 3 | Open My Articles. | Article listed with status DRAFT, tags Testing and QA, read time >= 1 min. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-ART-02: New article starts as DRAFT [Version : 1]<br>REQ-ART-10: My Articles [Version : 1] |
| Keywords: | None |

**Test Case US-17: Title length boundaries (0, 1, 200, 201 characters) [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Title length boundaries (0, 1, 200, 201 characters).

**Test design technique:** BVA

**Requirements:** REQ-ART-01

Preconditions:

Student A on My Articles > New Article.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Leave Title empty (0 chars), click Save Draft. | Error 'Please enter a title for your article'; nothing saved. |
| 2 | Title = 'A' (1 char), Save Draft. | Saved as draft. |
| 3 | Title = 200 x 'a', Save Draft. | Saved as draft. |
| 4 | Title = 201 x 'a', Save Draft. | Rejected with validation error (max 200); nothing saved. |
| 5 | Title = 5 spaces only. | Treated as empty: 'Please enter a title for your article'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-ART-01: Article input validation [Version : 1] |
| Keywords: | None |

**Test Case US-18: Excerpt length boundaries (500 / 501 characters) [Version : 1]**

| | |
|---|---|
| Author: | rawa |
| Summary: | |
| Excerpt length boundaries (500 / 501 characters).<br>**Test design technique:** BVA<br>**Requirements:** REQ-ART-01 | |
| Preconditions: | |
| Student A on New Article with valid title 'Excerpt test'. | |

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Leave Excerpt empty and save. | Saved (excerpt optional). |
| 2 | Excerpt = 499 characters, save. | Saved. |
| 3 | Excerpt = 500 characters, save. | Saved. |
| 4 | Excerpt = 501 characters, save. | Validation error (max 500); not saved. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-ART-01: Article input validation [Version : 1] |
| Keywords: | None |

| **Test Case US-19: Cover image URL validation [Version : 1]** | |
|---|---|
| Author: | rawa |
| Summary: | |
| Cover image URL validation.<br>**Test design technique:** EP (valid URL / invalid text / empty)<br>**Requirements:** REQ-ART-01 | |
| Preconditions: | |
| Student A on New Article, title 'Cover test'. | |

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Cover image = https://utfs.io/f/cover.png, save. | Saved; cover shown on article card. |
| 2 | Cover image = 'not a url', save via API article.create. | Validation error (invalid url); not saved. |
| 3 | Cover image empty, save. | Saved without cover (optional). |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-ART-01: Article input validation [Version : 1] |
| Keywords: | None |

| **Test Case US-20: Duplicate titles get unique slugs [Version : 1]** | |
|---|---|
| Author: | rawa |

Summary:

Duplicate titles get unique slugs.

**Test design technique:** EG, EP (new slug / existing slug)

**Requirements:** REQ-ART-03

Preconditions:

No article with slug 'software-testing-101' exists.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Create article 'Software Testing 101!' and publish. | URL is /articles/software-testing-101 (punctuation removed, lower-case, dashes). |
| 2 | Create a second article with the same title and publish. | URL is /articles/software-testing-101-1. |
| 3 | Create a third one. | URL is /articles/software-testing-101-2. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-ART-03: Unique URL slug [Version : 1] |
| Keywords: | None |

**Test Case US-21: Author edits own article; tags are replaced [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Author edits own article; tags are replaced.

**Test design technique:** Use case alternative flow

**Requirements:** REQ-ART-06, REQ-ART-03

Preconditions:

Student A has DRAFT 'Intro to Boundary Value Analysis' with tags Testing, QA.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Open My Articles > Edit; change title to 'BVA Explained', tags to 'Testing'; Save. | Toast 'Changes saved'. |
| 2 | Re-open the article. | Title 'BVA Explained', slug 'bva-explained', only tag Testing (QA removed). |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-ART-03: Unique URL slug [Version : 1]<br>REQ-ART-06: Edit own article [Version : 1] |
| Keywords: | None |

**Test Case US-22: User cannot edit another user's article [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

User cannot edit another user's article.

**Test design technique:** DT (actor = non-author), negative

**Requirements:** REQ-ART-06

Preconditions:

Student A authored 'BVA Explained'. Student B logged in.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | As Student B open /my-articles/<A's article id>/edit. | Edit not possible / redirected; no form for A's article. |
| 2 | Call article.update for that id with a new title as Student B. | FORBIDDEN 'You can only edit your own articles'; article unchanged. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-ART-06: Edit own article [Version : 1] |
| Keywords: | None |

## 1.5.Test Suite : TS-R5 Article Lifecycle (Publish / Archive / Delete)

State transition: DRAFT -> PUBLISHED -> ARCHIVED, delete from any state.

**Test Case US-23: Publish a draft article [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Publish a draft article.

**Test design technique:** ST (DRAFT -> PUBLISHED)

**Requirements:** REQ-ART-04

Preconditions:

Student A has DRAFT 'BVA Explained'.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | In My Articles click Publish on 'BVA Explained'. | Toast 'Article published!'; status PUBLISHED, published date = today. |
| 2 | Log out and open /articles. | 'BVA Explained' appears first in the list. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-ART-04: Publish workflow [Version : 1] |
| Keywords: | None |

**Test Case US-24: Only DRAFT articles can be published [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Only DRAFT articles can be published.

**Test design technique:** ST (invalid transitions PUBLISHED->PUBLISHED, ARCHIVED->PUBLISHED)

**Requirements:** REQ-ART-04

| | | |
|---|---|---|
| Preconditions:<br><br>Student A has one PUBLISHED and one ARCHIVED article. | | |
| #: | Step actions: | Expected Results: |
| 1 | Call article.publish on the PUBLISHED article. | BAD_REQUEST 'Only draft articles can be published'; publishedAt unchanged. |
| 2 | Call article.publish on the ARCHIVED article. | BAD_REQUEST 'Only draft articles can be published'; status stays ARCHIVED. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-ART-04: Publish workflow [Version : 1] | |
| Keywords: | None | |

| | | |
|---|---|---|
| **Test Case US-25: Archive own published article [Version : 1]** | | |
| Author: | rawa | |
| Summary:<br><br>Archive own published article.<br><br>**Test design technique:** ST (PUBLISHED -> ARCHIVED)<br><br>**Requirements:** REQ-ART-05, REQ-ART-07 | | |
| Preconditions:<br><br>'BVA Explained' is PUBLISHED by Student A. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open the article as Student A and choose Archive. | Toast 'Article archived'; status ARCHIVED. |
| 2 | Open /articles as a visitor. | 'BVA Explained' no longer listed. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-ART-05: Archive and delete own article [Version : 1]<br>REQ-ART-07: Public article listing [Version : 1] | |
| Keywords: | None | |

| | | |
|---|---|---|
| **Test Case US-26: Delete own article [Version : 1]** | | |
| Author: | rawa | |
| Summary:<br><br>Delete own article.<br><br>**Test design technique:** ST (any -> deleted)<br><br>**Requirements:** REQ-ART-05 | | |
| Preconditions:<br><br>Student A has article 'Delete me' (DRAFT). | | |
| #: | Step actions: | Expected Results: |

| | | |
|---|---|---|
| 1 | Open the article and choose Delete, confirm. | Toast 'Article deleted successfully'; redirected; article gone from My Articles. |
| 2 | Open its old URL /articles/delete-me. | 'Article not found'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-ART-05: Archive and delete own article [Version : 1] | |
| Keywords: | None | |

| **Test Case US-27: Non-author cannot publish, archive or delete [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Non-author cannot publish, archive or delete.<br><br>**Test design technique:** DT (actor x action), negative<br><br>**Requirements:** REQ-ART-04, REQ-ART-05 | | |
| Preconditions:<br><br>Student A authored DRAFT X and PUBLISHED Y. Student B logged in. | | |
| #: | Step actions: | Expected Results: |
| 1 | As Student B call article.publish on X. | FORBIDDEN 'You can only publish your own articles'. |
| 2 | As Student B call article.archive on Y. | FORBIDDEN 'You can only archive your own articles'. |
| 3 | As Student B call article.delete on Y. | FORBIDDEN 'You can only delete your own articles'; Y still exists. |
| 4 | As Student B open Y's page. | No Edit / Archive / Delete buttons shown. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-ART-04: Publish workflow [Version : 1]<br>REQ-ART-05: Archive and delete own article [Version : 1] | |
| Keywords: | None | |

## 1.6.Test Suite : TS-R6 Browse & Read Public Articles

/articles and /articles/[slug]. Techniques: EP, BVA on paging, DT on access.

| **Test Case US-28: Access control by status and viewer [Version : 1]** |
|---|
| Author: rawa |
| Summary:<br><br>Access control by status and viewer.<br><br>**Test design technique:** DT (status {PUBLISHED, DRAFT, ARCHIVED} x viewer {visitor, other student, author})<br><br>**Requirements:** REQ-ART-08 |
| Preconditions:<br><br>Student A has PUBLISHED P, DRAFT D, ARCHIVED R. |

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Logged out, open /articles/<P slug>. | Article P displayed. |
| 2 | Logged out, open /articles/<D slug> and /articles/<R slug>. | 'This article is not published' for both. |
| 3 | As Student B open D and R. | 'This article is not published'. |
| 4 | As Student A open D and R. | Both displayed with author controls. |
| 5 | Open /articles/no-such-slug. | 'Article not found'. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-ART-08: Article access control [Version : 1] | |
| Keywords: | None | |

| **Test Case US-29: Listing shows only published, newest first, with pagination [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary: | | |
| Listing shows only published, newest first, with pagination.<br>**Test design technique:** EP + BVA (13 articles vs page size 12)<br>**Requirements:** REQ-ART-07 | | |
| Preconditions: | | |
| Exactly 13 PUBLISHED articles (published on different times) and 2 DRAFT articles exist. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open /articles. | 12 articles on page 1, newest first; no DRAFT articles. |
| 2 | Go to page 2. | Exactly 1 article (the oldest). |
| 3 | Call article.getAllPublished with limit 0, 1, 50, 51. | 0 and 51 rejected by validation; 1 and 50 accepted. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-ART-07: Public article listing [Version : 1] | |
| Keywords: | None | |

| **Test Case US-30: Search and tag filter [Version : 1]** | |
|---|---|
| Author: | rawa |
| Summary: | |
| Search and tag filter.<br>**Test design technique:** EP (title match / excerpt match / no match / case)<br>**Requirements:** REQ-ART-07 | |

Preconditions:

Published 'BVA Explained' (excerpt 'boundary values', tag Testing) and 'Docker Basics' (tag DevOps).

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Type 'bva' in 'Search articles by title, content, or tags...'. | Only 'BVA Explained' shown (case-insensitive title match). |
| 2 | Search 'BOUNDARY'. | 'BVA Explained' shown (excerpt match). |
| 3 | Search 'zzzz'. | Empty result message. |
| 4 | Clear search, filter by tag Testing. | Only 'BVA Explained'. |
| 5 | Open the tag filter list. | Contains Testing and DevOps; tags used only by drafts are not listed. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-ART-07: Public article listing [Version : 1] |
| Keywords: | None |

**Test Case US-31: View counter increments only for published articles [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

View counter increments only for published articles.

**Test design technique:** EP (published / not published)

**Requirements:** REQ-ART-09

Preconditions:

'BVA Explained' PUBLISHED with 5 views; DRAFT D with 0 views.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Open 'BVA Explained' as a visitor. | View count becomes 6. |
| 2 | Refresh once. | View count becomes 7. |
| 3 | Author opens DRAFT D. | View count of D stays 0. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Low |
| Requirements | REQ-ART-09: View counter [Version : 1] |
| Keywords: | None |

**Test Case US-32: My Articles status filter [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

My Articles status filter.

**Test design technique:** EP (filter = all / DRAFT / PUBLISHED / ARCHIVED)

**Requirements:** REQ-ART-10

Preconditions:

Student A has 1 DRAFT, 2 PUBLISHED, 1 ARCHIVED; Student B has 1 PUBLISHED.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Open My Articles (all). | 4 articles of Student A only, most recently updated first. |
| 2 | Filter Draft / Published / Archived. | 1 / 2 / 1 articles respectively; Student B's article never shown. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Low |
| Requirements | REQ-ART-10: My Articles [Version : 1] |
| Keywords: | None |

## 1.7.Test Suite : TS-R7 Notification Centre (UC-S4-004)

Bell icon in the header. Techniques: BVA on badge (0, 1, 9, 10), ST (unread -> read).

**Test Case US-33: Unread badge boundaries (0, 1, 9, 10)** **[Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Unread badge boundaries (0, 1, 9, 10).

**Test design technique:** BVA

**Requirements:** REQ-NOT-02

Preconditions:

Student B logged in with 0 unread notifications.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Look at the bell. | No badge shown (0). |
| 2 | Generate 1 notification (Student A invites B to a course). Refresh. | Badge shows 1. |
| 3 | Generate notifications until B has 9 unread. | Badge shows 9. |
| 4 | Generate the 10th. | Badge shows '9+'. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-NOT-02: Unread badge [Version : 1] |
| Keywords: | None |

**Test Case US-34: Dropdown lists 20 most recent, newest first** **[Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Dropdown lists 20 most recent, newest first.

**Test design technique:** BVA (20 / 21 notifications), EP (empty)

**Requirements:** REQ-NOT-03

Preconditions:

Student B has 0 notifications.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Open the bell. | 'No notifications yet'. |
| 2 | Create 21 notifications for B, open the bell. | Exactly 20 shown, newest at top; oldest (1st) not shown. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-NOT-03: Recent notifications dropdown [Version : 1] |
| Keywords: | None |

**Test Case US-35: Click invitation notification marks read and opens Pending Invitations [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Click invitation notification marks read and opens Pending Invitations.

**Test design technique:** ST (unread -> read), use case

**Requirements:** REQ-NOT-04, REQ-SHR-04

Preconditions:

Student B has 3 unread, one of them 'New course invitation'.

| #: | Step actions: | Expected Results: |
|---|---|---|
| 1 | Open the bell and click 'New course invitation'. | Navigates to /settings?tab=invitations showing the invitation. |
| 2 | Look at the bell. | Badge decreased from 3 to 2; the clicked item is shown as read. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | High |
| Requirements | REQ-SHR-04: Invitee accepts or declines a pending invitation [Version : 1]<br>REQ-NOT-04: Mark one as read [Version : 1] |
| Keywords: | None |

**Test Case US-36: Mark all as read [Version : 1]**

| | |
|---|---|
| Author: | rawa |

Summary:

Mark all as read.

**Test design technique:** ST (all unread -> read)

**Requirements:** REQ-NOT-05

Preconditions:

Student B has 5 unread notifications.

| #: | Step actions: | Expected Results: |
|---|---|---|

| | | |
|---|---|---|
| 1 | Open the bell and click 'Mark all as read'. | All items shown as read; badge disappears. |
| 2 | Reload the page. | Still no badge (persisted). |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-NOT-05: Mark all as read [Version : 1] | |
| Keywords: | None | |

| **Test Case US-37: User cannot read or mark another user's notification [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>User cannot read or mark another user's notification.<br><br>**Test design technique:** EG, security negative<br><br>**Requirements:** REQ-NOT-06 | | |
| Preconditions:<br><br>Student A has notification N (id known). Student B logged in. | | |
| #: | Step actions: | Expected Results: |
| 1 | As Student B open the bell. | N is not listed; unread count excludes A's notifications. |
| 2 | As Student B call notification.markAsRead with N's id. | Error 'Notification not found'; N stays unread for Student A. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | High | |
| Requirements | REQ-NOT-06: Notification privacy [Version : 1] | |
| Keywords: | None | |

| **Test Case US-38: View all notifications page with paging [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>View all notifications page with paging.<br><br>**Test design technique:** EP + BVA (limit 0, 1, 100, 101)<br><br>**Requirements:** REQ-NOT-09 | | |
| Preconditions:<br><br>Student B has 25 notifications. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open the bell and click 'View all'. | Full notification page opens (/notifications) showing the first 20 and a way to load more. |
| 2 | Load more. | Remaining 5 shown; no duplicates. |

| | | |
|---|---|---|
| 3 | Call notification.getAll with limit 0, 1, 100, 101. | 0 and 101 rejected; 1 and 100 accepted. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-NOT-09: Full notification history [Version : 1] | |
| Keywords: | None | |

## 1.8.Test Suite : TS-R8 Notification Preferences

Settings > Notification Preferences. Techniques: EP, BVA (5..30 minutes), DT (channel x type).

| **Test Case US-39: Default preferences are created on first visit [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Default preferences are created on first visit.<br><br>**Test design technique:** EP (new user without preferences)<br><br>**Requirements:** REQ-NOT-07 | | |
| Preconditions:<br><br>Newly approved Student E who never opened Settings. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open Settings > Notification Preferences. | Page loads with default toggles; reminder timing shows 10 minutes. |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-NOT-07: Notification preferences [Version : 1] | |
| Keywords: | None | |

| **Test Case US-40: Save in-app / email preferences per type [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Save in-app / email preferences per type.<br><br>**Test design technique:** DT (2 channels x 4 types)<br><br>**Requirements:** REQ-NOT-07 | | |
| Preconditions:<br><br>Student B on Notification Preferences. | | |
| #: | Step actions: | Expected Results: |
| 1 | Turn OFF email 'Collaboration invites' and in-app 'Audit log alerts', keep the rest ON; Save. | Success message. |
| 2 | Reload page. | Exactly those two toggles are OFF. |
| 3 | Student A invites Student B to a course. | In-app notification appears; no invitation email is sent to Student B. |

| | |
|---|---|
| Execution type: | Manual |
| Estimated exec. duration (min): | |
| Importance: | Medium |
| Requirements | REQ-NOT-07: Notification preferences [Version : 1] |
| Keywords: | None |

| **Test Case US-41: Class reminder timing boundaries [Version : 1]** | | |
|---|---|---|
| Author: | rawa | |
| Summary:<br><br>Class reminder timing boundaries.<br><br>**Test design technique:** BVA (4, 5, 30, 31 minutes)<br><br>**Requirements:** REQ-NOT-08 | | |
| Preconditions:<br><br>Student B on Notification Preferences. | | |
| #: | Step actions: | Expected Results: |
| 1 | Open 'Reminder Timing' dropdown. | Options 5, 10, 15, 30 minutes only. |
| 2 | Select 5 minutes, save, reload. | 5 minutes persisted. |
| 3 | Select 30 minutes, save, reload. | 30 minutes persisted. |
| 4 | Call notification.updatePreferences with classReminderMinutes 4 and 31. | Both rejected by validation; stored value unchanged (30). |
| Execution type: | Manual | |
| Estimated exec. duration (min): | | |
| Importance: | Medium | |
| Requirements | REQ-NOT-08: Reminder timing range [Version : 1] | |
| Keywords: | None | |